

Activity and Recovery Among Youth With Concussion: A Meta-analysis

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abstract

CONTEXT: Concussions in youth are highly prevalent. Previously, rest was prescribed to prevent adverse outcomes; however, early return to activity is emerging to improve the recovery trajectory.

OBJECTIVES: To evaluate the effectiveness of early return to physical and social activity interventions on recovery outcomes in youth with concussion.

DATA SOURCES: A systematic review was conducted up to October 2022.

STUDY SELECTION: We included randomized controlled trials (RCTs) and non-RCTs that reported effects of activity-based interventions on symptoms, quality-of-life (QoL), and return to preinjury activity levels in children and youth after a concussion.

DATA EXTRACTION: Three authors independently extracted data on publication year and country, study setting and design, sample size, participant demographics, intervention, outcome(s), and author conclusion. Meta-analysis was conducted on appropriate RCTs.

RESULTS: Twenty-four studies were included in the final review, of which 10 were RCTs. There was a significant effect of activity interventions on symptom reporting (standardized mean difference, 0.39 [95% confidence interval, 0.15 to 0.63]; I^2 , 0%; P = .002). There was not a significant effect of activity-based interventions on QoL (mean difference, −0.91 [95% confidence interval, −7.76 to 5.94]; I^2 , 0%; P = .79). No meta-analysis was performed on return to preinjury activity levels because of insufficient number of RCTs conducted.

LIMITATIONS: One outcome was excluded from the meta-analysis. Interventions emphasizing social activity were lacking.

CONCLUSIONS: Findings indicate that activity-based interventions may significantly improve concussion symptoms. There is insufficient data to understand the effect of activity-based intervention on QoL and return to preinjury activity levels.



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Ms Chauhan conceptualized, designed, and undertook the review, drafted the initial manuscript, and reviewed and revised the manuscript; Ms Cheng designed the review, screened for inclusion of studies, conducted quality ratings, and extracted data; Ms Tsow screened for inclusion of studies, conducted quality ratings, and extracted data; Drs Sakakibara and Babul contributed to the conceptualization, design, and analysis of data, and reviewed and revised the manuscript; Dr Schmidt conceptualized, designed, and supervised the review, and reviewed and revised the manuscript; and all authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

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Concussions are highly prevalent brain injuries caused when a sudden force to an individual's head or body results in a back-and-forth shaking motion of the brain.¹ Between 2003 and 2013, there was an observed increase in the prevalence of concussions compared with previous years.² Youth particularly experience a high prevalence of concussion because of the high levels of activity they undertake on a daily basis.³ In addition, youth have a slower recovery rate and negative prolonged effects compared with adults.³ This, in association with their developing brain, makes it even more important to recognize a concussive event and understand prevention and recovery methods in youth after a concussion.

The symptoms of a concussion can greatly differ in terms of severity, type, duration, and between and within individuals across concussive events.⁴ Types of symptoms can be described in 4 categories: emotional, physical, cognitive, and sleep-related.⁵ Emotional symptoms come in the form of anxiety, depression, and lability, for example.⁶ Physical symptoms are commonly presented as headaches and dizziness, among other things, whereas cognitive symptoms include memory impairments and attention deficits.^{5,6} Lastly, symptoms relating to sleep are prevalent in the forms of insomnia or daytime hypersomnia.⁷ Many individuals will experience resolution of symptoms within 4 weeks of the injury event.^{8,9} Approximately 30% of individuals will experience symptoms lasting >3 months, which are known as persistent postconcussive symptoms.^{8–11}

Current guidelines suggest that an earlier return to activity (ie, after 24–48 hours of concussion) can improve recovery because of physiologic and psychological benefits.¹² Progression to moderate physical activity levels correlate

with a positive recovery trajectory, because physical activity can positively affect brain recovery.^{13,14} Some studies have found that social connection is also important in concussion recovery, particularly in a youth population.^{15,16} A lack of social connection has a significant negative effect on outcomes in youth with concussion, evidenced by a reduction in hobbies, limited engagement in school activities, and increased emotional symptoms.¹⁵

Although there are individual studies demonstrating the effect of physical and social activities on recovery after concussion, there has not been any systematic pooling of evidence to understand the influence and effectiveness of activity in youth. Therefore, this study aimed to pool together data to better understand the effects of activity after a concussion in youth. The objectives of this systematic review with meta-analysis were to:

1. examine the influence of physical and social activity on recovery in youth with concussions; and
2. understand the effectiveness of interventions that promote physical and/or social activity on recovery in youth with concussion.

METHODS

Study Design and Registration

A systematic review with meta-analysis was undertaken. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines were followed. The protocol was registered with PROSPERO in April 2020 (#CRD42020176278).

Information Sources

The search strategy included the following databases: Medline Ovid, Embase Ovid, CINAHL EBSCO, and PsychINFO. The references of included studies were reviewed to

ensure maximum inclusion of articles. The main search terms included concussion, exercise, social participation, child, and adolescents (full search strategy can be found in the Supplemental Information). Search strategy and terms were generated with a librarian at the University of British Columbia.

Inclusion and Exclusion Criteria

Participants

Included studies were composed of children and youth living in the community, aged between 0 and 18 years, and who had sustained at least 1 concussion. Studies including mixed populations (ie, adults and children) were included if >50% of the sample size was within the nominated age range of 0 to 18 years. Inclusion criteria did not consider time because injury as both acute and chronic studies were deemed important for this review to understand the full scope of potential benefits and risks.

Studies

All quantitative studies with data on intervention outcomes (eg, randomized controlled trials [RCTs], feasibility studies, retrospective studies, and others) were considered for review. Studies only published in English and in peer-reviewed journals were included. Studies were excluded if they used a review methodology (literature, scoping, systematic) or were gray literature (eg, thesis, conference abstracts).

Intervention

Activity-based interventions, including physical and social activities, after concussion in youth were considered. Physical activity was defined as any aerobic or anaerobic activity. Social activities were defined as activities that engage an individual on the basis of their roles, social setting, and preferences/interest (eg,

involvement in clubs, interactions with peers). Studies including a combination of physical and social activity components were also included.

Studies were included if they had no comparison group; comparisons to participants with a concussion who underwent usual care intervention, such as commonly prescribed treatment in a particular context; or comparisons between 2 activity-based interventions.

Outcomes

The primary outcome was recovery after concussion. We included studies for review if they reported on any of 3 main outcomes related to recovery: including

1. symptom reporting, including duration, severity, and number of symptoms;
2. quality of life (QoL); and
3. days until return to preinjury activity.

Search Protocol

Citations were imported into an online reference manager, Covidence. After the removal of duplicate articles, each article was independently screened by 2 authors (R.C., A.C., R.T.) on the basis of relevancy by title, and abstract, and then a full text review was completed. Discrepancies among authors were resolved through discussion and the addition of a third author, as needed.

Data Extraction

The full text of all selected articles was gathered for data extraction and synthesis. When needed, authors of the original study were contacted to obtain data which were not reported in the article. Data extraction was conducted by 3 authors and followed a planned extraction document which was

developed before extraction. The data extracted included year of the study, authors, setting, study design, participant demographic data, number of participants throughout the study, information on the intervention, comparator information, recovery outcome measures, and results of the intervention.

Quality of Study

The quality of included studies was assessed through risk of bias assessment, utilizing 2 assessors. The Cochrane Risk of Bias Tool was used for RCTs. The Cochrane Risk of Bias tool assesses various domains, including randomization-related biases (ie, allocation, sequence), intervention-related biases (blinding), incomplete outcome data biases, outcome measurement biases, and bias related to selective reporting of outcomes.¹⁷ These domains were examined together and gradings were determined according to the Cochrane Risk of Bias tool, including low risk of bias, high risk of bias, or unclear. For example, if all domains clearly reported a low risk of bias, the overall risk of bias grade would be low.¹⁷ However, if 1 domain reported a high risk of bias, the overall risk of bias grade for the article would be deemed high.¹⁷

To assess non-RCTs, the Risk of Bias in Nonrandomized Studies of Intervention (ROBINS-I) assessment tool was used. The ROBINS-I assesses bias through 7 domains, including preintervention biases (ie, confounding, participant selection), during intervention biases (ie, intervention classification), and postintervention biases (ie, intervention deviations, missing data, outcome measurements, selective reporting of outcomes).¹⁸ These domains were assessed through “signaling questions,” which were answered as 1 of the

following: “yes,” “probably yes,” “probably no,” “no,” or “no information”¹⁸. Once domain-specific biases were established, an overall judgment grade was determined as “low,” “moderate,” “serious,” or “critical” risk.¹⁸ For example, the risk of bias was assessed as low if none of the domains show a potential for bias, unlikely with non-RCTs.¹⁸

Descriptive Analysis

Descriptive analysis followed a narrative approach. Data from included studies addressing the 3 main outcomes of interest were tabulated and compiled. Particular attention was given to describing the intervention and the effect of the intervention on the outcome of interest. The questionnaires and measures used to test the outcomes were specified, along with the number of studies showing significant effects of the intervention on an outcome. Additionally, sample size, along with the number of males and females, was described.

Meta-Analysis

A meta-analysis was completed on data from RCTs with the same outcome in 2 or more studies with suitable heterogeneity for a combined analysis. Clinical, methodological, and statistical heterogeneity were assessed a priori to see whether studies could be accurately combined. Clinical heterogeneity was defined as differences in participants, interventions, and outcome types.¹⁹ Methodological heterogeneity looked at differences in the study design, the outcome measurements used, and risk of bias.¹⁹ Lastly, statistical heterogeneity assesses any differences in the effects seen because of the intervention and was assessed by utilizing the I^2 statistic.¹⁹ Studies with symptom reporting, QoL, and days to return to activity outcomes were included

in the meta-analysis. Plots and values were expressed using either mean differences (MDs), if the same outcome was used in all studies, or standardized MDs, if different outcomes were used among studies in the meta-analysis to account for differences in measurement tools.¹⁹ Random-effect versus fixed-effect methods were chosen on the basis of the presence or absence of heterogeneity among studies. A random-effect model was chosen if intervention variability was seen across studies and there was enough heterogeneity, for which a random-effect model would have given more weight to studies with a smaller sample size.¹⁹

Sample size, baseline, and postintervention measures such as mean, SD, and confidence intervals (CIs) were gathered from the studies during the data extraction stage. Authors were contacted to retrieve data in the absence of enough data available to calculate the mean or SD. Postintervention data were inputted into RevMan 5.4 and used in the analysis, as opposed to change scores (postintervention minus baseline), because the postintervention score is similar because of the randomized control methodology.¹⁹ Postintervention data were used as change scores for all outcomes that were noted calculable.

RESULTS

There were 5642 studies initially identified in the search after removal of duplicates (Fig 1). A total of 5618 studies were excluded because they did not fit the predefined criteria. After screening, 24 studies were included in the final review, with 10 RCTs, 4 case series/studies, 4 retrospective study designs, 3 quasi-experimental designs, 2 pre-post studies, and 1 feasibility study. The sample size of individual studies ranged from 6 to

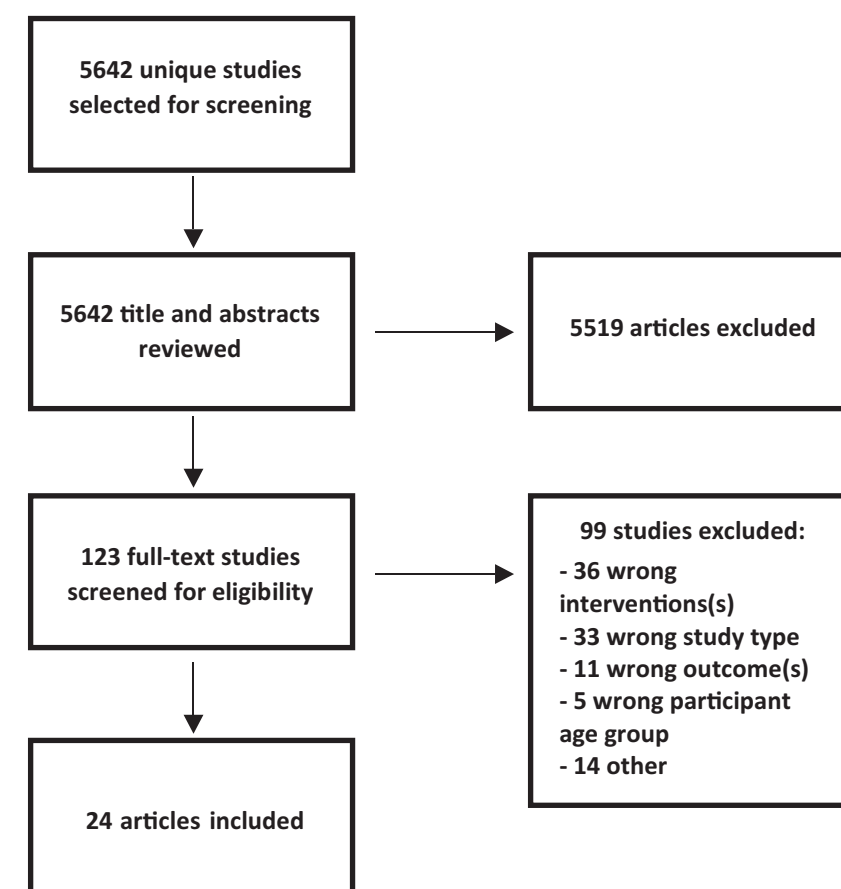


FIGURE 1
Study selection process of included and excluded articles.

677 (Table 1). Four studies included adult populations (>18 years), with the maximum age being in the mid-20s. The condition of having >50% of individuals within the nominated age range of 0 to 18 years was still met. Initiation of intervention after the injury event varied among studies.

All studies included an element of physical activity in their intervention protocol. No studies included interventions composed of social activities. The most prominent type of physical activity intervention was aerobic exercise, with 17 studies including it as at least 1 element in the intervention. One study focused on mindfulness within yoga, which occurred in-person.²⁰ In 1 retrospective study, individuals

participated in a broad array of noncontact exercises.²¹

Most studies occurred within a concussion/sports medicine clinic ($n = 10$). Studies included either monitored in-clinic programs ($n = 4$), at-home exercise programs ($n = 7$), or a combination of both ($n = 7$). In 3 studies, participants were provided a choice to exercise supervised at home or a gym; in 1 study, individuals engaged in 1 self-prescribed activity or physician-prescribed activity only, and in 2 studies, the location of exercise was not explicitly stated. One study involved a virtual or telerehabilitation implementation method, in which weekly video meetings were scheduled.²²

TABLE 1 Description of Studies Included in the Review

Source	Quality Rating	Study Design	Population		Intervention Group	Control Group	Outcomes	Results
			Intervention	Control				
Anderson et al, 2018 ²³	Cochrane RoB: high	RCT	<i>n</i> = 30; mean age, 13.9 y; age range, ^a 10–18 y; females, 5; males, 25	<i>n</i> = 30; mean age, 13 y; females, 14; males, 16	A 2-min warmup activity followed by submaximal exercise for 6 min. For the submaximal exercise, resistance was increased and determined on the basis of participants' height. Frequency, 1 exercise session	N/A	CogSport Symptom Scale	Exercise-reduced symptom expression in youth with postconcussive symptoms
Bailey et al, 2019 ²⁴	Cochrane RoB: moderate	Pilot RCT	<i>n</i> = 7; For the entire sample (intervention + control): mean age (SD), 15.75 (1.39) y; age range, 14–18 y; females, 44%; males, 56%	<i>n</i> = 8	A monitored exercise routine with in-clinic and at-home workout sessions. Frequency, daily for 6 wk (in-clinic and at home); duration, 20 min; intensity, 80% of the participants' HR threshold	Light activity. Frequency, daily (3 wk of stretching, 3 wk of walking); duration, 5 stretching activities, followed by 3 wk of 20 min walks	PCS-R; BDI-II	Exercise had a positive effect on symptoms, controlling for depression
Chan et al, 2018 ²⁵	Cochrane RoB: high	RCT	<i>n</i> = 10; mean age (SD), 15.9 (1.66) y; age range, 12–18 y; females, 6; males, 4	<i>n</i> = 9; mean age (SD), 15.1 (1.42) y; females, 8; males, 1	Participants received treatment as usual, followed by physiotherapists providing rehabilitation consisting of submaximal aerobic training, light coordination and sport-specific exercises, and visualization and imagery techniques. Physiotherapists devised a home exercise program for the participant. Frequency, weekly visits until the participant could independently do the home program	Treatment as usual Before randomization: 1. Education about concussions 2. Return to school consultation 3. Psychiatrist consultation; medicine + referrals After randomization: Mental health and medicine follow-ups, if needed	PCSS; BDI; PedsQL multidimensional fatigue scale, teen report standard version; BESS; IMPACT	The intervention had a statistically significant effect on postconcussion symptoms.
Chrisman et al, 2017 ²⁶	ROBINS-I: moderate	Pilot study retrospective chart review	<i>n</i> = 83; mean age (SD), 14.9 (2.3) y; age group, youth only (<18 y); females, 54%	N/A	At-home exercise program before a follow-up. Frequency, follow-up visit in 1–2 wk. Intervention continued until participant could complete the treadmill test without worsening symptoms. Duration, for as long as physical activity was completed on the initial treadmill test with a maximum limit 20 min; intensity, 80% of HR threshold-producing symptoms, as tested	N/A	SCAT-2, symptom subscale	The exercise intervention was associated with improved concussion symptoms, even for long after the concussive event.

TABLE 1 Continued

Source	Quality Rating	Study Design	Population			Intervention Group	Control Group	Outcomes	Results
			Intervention	Control					
Chrisman et al, 2019 ²⁷	Cochrane RoB: high	Pilot RCT	$n = 19$; mean age (SD), 15.9 (1.66) y; age range, 12–18 y	$n = 11$; mean age (SD), 15.1 (1.42) y		A 6-wk home aerobic exercise program according to the participants' exercise preferences and lifestyle. Frequency, daily for 6 wk; duration, 5–10 min greater than moderate—vigorous physical activity measured; intensity, 80% of HR threshold-producing symptoms, as tested	Home stretching program completed for 5–10 min daily	Symptoms, the Health Behavior Inventory; PedsQL parent-reported and child-reported; change in physical activity, accelerometry; fear avoidance, Fear of Pain Questionnaire	The intervention group showed improved QoL and symptom outcomes.
Chrisman et al, 2021 ²⁸	ROBINS-I: moderate	Feasibility	$n = 19$; mean age (SD), 14.3 (2.2) y; age range, 10–20 y; females, 79%; males, 21%			Home aerobic exercise program (mobile subthreshold exercise program). Frequency, daily; duration, 6 wk; intensity, started at 10 min of activity at HR of 120 and progressed weekly to a max of 60 min of activity at a HR of 140	N/A	Symptoms, Health Behavior Inventory; PedsQL: Fear of Pain Questionnaire, adapted for concussive symptoms; Patient Health Questionnaire-9; Generalized Anxiety Disorder Scale-7; Adolescent Sleep Wake Scale-10 item	Concussion symptoms showed significant improvements between the time points. Health-related QoL saw improvements over the 6 wk.
Dobney et al, 2017 ²⁹	ROBINS-I: moderate	Pre-post study	$n = 277$; mean age (SD), 14.3 (2.3) y; age group, youth only (<18 y); female, 135; male, 142	N/A		Physiotherapist-led rehabilitation which consisted of: 1. Aerobic activity. Activity type, stationary bike or treadmill; duration, 15 min; intensity, 60% of max HR (age-predicted) 2. Coordination/skill practice. Activity type, personalized to participants' favored sport/activity; duration, max 10 min 3. Visualization. Activity type, sport/activity imagery based on participants' preference; duration, 5–10 min 4. Education and motivation. Frequency, at each clinic visit. An at-home program with the above steps was developed to be completed daily.	POSS		Postintervention data showed that there was a decrease in symptoms.
Dobney et al, 2018 ³⁰	ROBINS-I: moderate	Case series	$n = 677$; mean age (SD), 14.3 (2.3) y; age range,	N/A		Physiotherapist-led rehabilitation which consisted of:	N/A	PCS R	Significant decrease in symptoms after intervention

TABLE 1 Continued

Source	Quality Rating	Study Design	Population		Control Group	Outcomes	Results
			Intervention	Control			
Gagnon et al, 2009 ³¹	ROBINS-I: moderate	Case studies	<i>n</i> = 16; mean age, 14.25 y; age range, 10–17 y; female: 5; male, 11	N/A	N/A	POS-R	Recovery trajectory improved across all participants
					Active rehabilitation which consisted of: 1. Submaximal aerobic training. Activity type, stationary bike or treadmill; duration, max 15 min; intensity, 50%–60% of max capacity of HR 2. Light coordination exercises. Activity type, personalized to participants' preferred or main sport; duration, max 10 min 3. Visualization and positive imagery techniques. Activity type, sport specific. 4. Home exercise program. Activity type, submaximal aerobic training and coordination exercises based on the participants' activities; frequency, daily until symptom free for 1 wk; duration, symptom tolerance levels as noted during monitored sessions		(independent of when rehabilitation was initiated).

TABLE 1 Continued

Source	Quality Rating	Study Design	Population				Intervention Group	Control Group	Outcomes	Results
			Intervention	Control						
Gagnon et al, 2016 ⁸²	ROBINS-I: low	Case series	<i>n</i> = 10; age range, 14–18 y; female, 3; male, 7	N/A			Active rehabilitation which consisted of (steps 1–3 performed under physiotherapist supervision): 1. Submaximal aerobic training. Activity type, stationary bike or treadmill; duration, max 15 min; intensity, 60% of max capacity of HR 2. Light coordination exercises. Activity type, personalized to participants' preferred or main sport; duration, max 10 min 3. Visualization and positive imagery techniques. Activity type, sport-specific motor skill 4. Home exercise program. Activity type, includes all components of the intervention; frequency, daily until symptom-free for 1 wk; duration, 20–30 min Duration and intensity of at-home exercises were adjusted according to the participants' symptoms and concussion experience.	N/A	PCSS; BDI-II; PedsQL multidimensional fatigue scale; balance and coordination; imPACT	Reduced symptoms were observed in all 10 cases.
Gauvin-Lepage et al, 2020 ⁸³	ROBINS-I: serious	Quasi-experimental control group design	<i>n</i> = 36; mean age (SD), (1.9) y; age range, 8–17 y; females, 21 (58.4%); males, 15 (41.6%)	<i>n</i> = 13; mean age (SD), 13.2 (2.6) y; females, 5 (38.5%); males, 8 (61.5%)			Standard care + active rehabilitation intervention. Frequency, daily. 1. Aerobic activity. Duration, 15 min; intensity, max. HR corresponding to 50%–60% of their maximal capacity 2. Coordination/sport-specific activity. Duration, 10 min 3. Mental imagery. Duration, 5 min 4. Education 5. Home program	Standard care: Rest and/or restricted activities based on symptoms; general education; academic adaptations; gradual return to school; vigorous physical activity restrictions until complete symptom resolution	POS; PedsQL; mood and anxiety; fatigue; balance and coordination; physical activity participation; cognitive function	There was a decrease in symptoms in both the intervention and control groups. The intervention group did not show significant differences compared with the control group. For QoL, individuals in the intervention group showed higher QoL scores when compared with the control group.

TABLE 1 Continued

Source	Quality Rating	Study Design	Population			Intervention Group	Control Group	Outcomes	Results
			Intervention	Control					
Gladstone et al, 2019 ³⁴	Cochrane RoB: high	Partially masked exploratory RCT	<i>n</i> = 15; mean age (SD), 15.22 (1.37) y; age range, 12–17 y; female, 10; male, 5	<i>n</i> = 15; mean age (SD), 15.50 (1.80) y; female, 7; male, 8		At-home personalized exercise program. Frequency, 5–6 d a wk; duration, 80% of symptom-free HR, as tested	Stretching program, with new stretches given every 2 wk. Frequency, 5–6 d a wk	PedsQL. Parent-reported and child-reported; cognition, the National Institutes of Health Toolbox Cognition Battery	QoL improvements were seen in both the intervention and control groups.
Grabowski et al, 2017 ³⁵	ROBINS-I: serious	Retro-spective study	<i>n</i> = 25; mean age, 15.16 y; age range: 12–20 y. Participants were instructed to complete 1, 2, or all 3 of the rehabilitation components on the basis of their symptoms + type of PCS. Treatment <i>n</i> : Np, 6; Nc, 3; Nv, 5; Npc, 3; Npv, 3; Nov, 3; Nail, 2. Male/female: Np, 4 m/2 f; Nc, 0 m/3 f; Nv, 4 m/1 f; Npc, 0 m/3 f; Npv, 2 m/1 f; Nov, 0 m/3 f; Nail, 1 m/1 f	N/A		Physiotherapist-supervised rehabilitation consisting of up to 3 components: 1. Cardiovascular exercise protocol. Activity type, participants preference (ie, walking, jogging); frequency, 5–6 d a wk; intensity, 80% of symptom-free HR 2. Vestibular/oculomotor therapeutic exercise. Activity type, training and habituation exercises; frequency, daily 3. Cervicothoracic manual therapy and therapeutic exercise. Activity type, joint mobilization, strength and flexibility exercises; frequency, daily After progression of the program, activities were introduced on the basis of particular sports.	N/A	PCSS; BESS	Improvements in PCSS
Howell et al, 2022 ³⁶	Cochrane RoB: high	Block stratified pilot randomized clinical trial	<i>n</i> = 9; mean age, 14.2 (2.1) y; age range, 10–18 y; females, 5 (56%); males: 4 (44%)	<i>n</i> = 7; mean age, 13.6 (1.7) y; females, 4 (57%); males, 3 (43%)		Individualized early aerobic exercise. Frequency, 5 times a wk; duration, 20 min per d; intensity, 80% of HR at the end of the exercise test	Health care provider instructions. Typically involved gradual progression of physical activity without exceeding symptom exacerbation thresholds.	PCSI	The intervention group performed a higher volume of exercise per wk and developed less persistent postconcussion symptoms 28 d after the injury. There was no significant difference for PCSI scores found between the intervention and control groups from baseline to follow-up. Possible benefits of the aerobic activity intervention
Kurowski et al, 2017 ³⁷	Cochrane RoB: high	Pilot RCT	<i>n</i> = 12; mean age (SD), 15.22 (1.37) y; age range, 12–17 y; females, 10; males, 5	<i>n</i> = 14; mean age (SD), 15.50 (1.80) y; females, 7; males, 8		Subsymptom exacerbation aerobic training. Activity type, portable exercise bike; frequency, 5–6 d a wk; duration: 80% of symptom exacerbation, as tested weekly	Full-body stretching program with new stretches every 2 wk. Frequency, 5–6 d a wk for at least 6 wk	PCSI, self-reported and parent/guardian-reported	Possible benefits of the aerobic activity intervention

TABLE 1 Continued

Source	Quality Rating	Study Design	Population		Intervention Group	Control Group	Outcomes	Results
			Intervention	Control				
Lawrence et al, 2018 ³⁸	ROBINS-I: moderate	Retro-spective study	<i>n</i> = 253; median age 17 (IQR, 15.0–20.0) y; females, 105; males, 148	N/A	Self-initiated (jogging, running, swimming, cycling, or stationary equipment) or physician-prescribed (graded return to activity guideline) aerobic activity	N/A	Time to full return to sports, work, or school (assessed in d from injury event)	Returning to activity sooner was associated with better outcomes. A faster return to full activity was related to quicker recovery.
Leddy et al, 2019 ⁴⁰	Cochrane RoB: high	Parallel RCT	<i>n</i> = 52; mean age (SD), 15.3 (1.6) y; age range, 13–18 y; females, 24; males, 28	<i>n</i> = 51; mean age (SD), 15.4 (1.7) y; females, 24; males, 27	An aerobic exercise intervention. Activity type, stationary bike or treadmill at home or a gym setting (walking or jogging allowed if exercise equipment was not available); frequency, daily for 30 d. Those experiencing delayed recovery were given further treatment. Duration, 20 min; intensity, 80% of HR causing exacerbated symptoms	Stretching program with weekly advances. Frequency, daily; duration, 20 min	PCSS	Participation in the activity intervention was associated with a quicker recovery.
Leddy et al, 2019 ⁴⁰	ROBINS-I: serious	Quasi-experimental design, comparison of 2 convenience samples	<i>n</i> = 24; mean age (SD), 15.13 (1.42) y; age range, 13–18 y; males, 24	<i>n</i> = 30; mean age (SD), 15.33 (1.40) y; males, 30	An aerobic exercise intervention. Activity type, stationary bike or treadmill at home or a gym setting; frequency, daily; duration, 20 min with a 5 min warmup and 5–10 min cool-down; intensity, 80% of HR causing exacerbated symptoms	Resting program with instructions not to participate in sports or physical activity	PCSS	Engaging in the exercise intervention was associated with prolonged recovery.
Leddy et al, 2021 ⁴¹	Cochrane RoB: high	RCT	<i>n</i> = 61; mean age, 15.5 (1.4) y; age range, 13–18 y; female, 23 (38%); male, 38 (62%)	<i>n</i> = 57; mean age, 15.9 (1.4) y; female, 21 (37%); male, 36 (63%)	Aerobic exercise, home exercise. Intensity, 90% of max HR during first BCGT visit; duration, 20 min; frequency, daily	Stretching exercise. A combination of progressive light stretches and breathing exercises which would not increase HR	PCSI	There was a significant effect of aerobic exercise on recovery within the first 4 wk. Median d until recovery was faster in the intervention group (14 d versus 19 d).
Micay et al, 2018 ⁴²	Cochrane RoB: high	Pilot RCT	<i>n</i> = 8; mean age (SD), 15.8 (1.2) y; age range, 14–18 y; males, 8	<i>n</i> = 7; mean age (SD), 15.6 (1.0) y; males, 7	An aerobic exercise intervention using a stationary bike. Frequency, 8 sessions, with exercise for 2 d, followed by 1 d rest. Intensity and duration varied depending on age and recovery	Usual care: 6-stage progression of activity (Berlin 2016 guidelines)	D until medical clearance; PCSS	Significant effect of the intervention on symptom severity between groups, with the intervention group showing decreased symptoms

TABLE 1 Continued

Source	Quality Rating	Study Design	Population		Intervention Group	Control Group	Outcomes	Results
			Intervention	Control				
Paniccia et al, 2019 ²⁰	ROBINS-I: serious	Case series	<i>n</i> = 6; age range, 13–17 y; females, 4; males, 2	N/A	stage, but all sessions included a 5 min warmup and 5 min cool-down. Session 1: duration, 10 min; intensity, 50% of participants' age-predicted maximum HR. Session 2: duration, 20 min; intensity, same as session 1. Sessions were all 20 min, with the exercise intensity increasing by 5% until 70% of the participants' age-predicted maximum HR was achieved. Intensity at 70% stayed for the final 3 sessions.	N/A	Godin Leisure-Time Exercise Questionnaire; PCS; HR variability	Symptom improvement was reported. No significant effects of intervention on daily activity engagement and enjoyment
Shore et al, 2022 ²²	ROBINS-I: moderate	Precase–postcase series design	<i>n</i> = 3; mean age, 15 y; age range, 14–17 y; females, 2 (66.66%); male(s), 1 (33.34%)	N/A	Tele-active rehabilitation program for 6 wk with a kinesiologist. Home program, 3 d per wk; kinesiologist meeting, once per wk 1. Progressive subsymptom threshold aerobic exercise. Duration, initially 10 min and progresses to a max of 30 min 2. Sport coordination drills. Duration, 10 min 3. Education and support	N/A	PCS; Brief Illness Perception Questionnaire; COPM; satisfaction	Symptom reduction was seen in all 3 participants from preintervention to postintervention, with the greatest difference being seen between wk 0 and wk 3. An improvement in performance (school, sports, hobbies) was seen.
Stumph et al, 2020 ²¹	ROBINS-I: moderate	Retrospective study	<i>n</i> , 112. Age groups breakdown: <10 y, 0; 10–12 y, 29; 13–15 y, 62; 16–17 y, 20; ≥18 y, 1. Females, 54; males, 58	<i>n</i> = 75. Age groups breakdown: <10 y, 1; 10–12 y, 13; 13–15 y, 41; 16–17 y, 17; ≥18 y, 3. Females, 30; males, 45	Subsymptom exacerbation exercise. Activity type, noncontact; intensity, not standardized across physicians	Physical activity avoidance until asymptomatic	D until symptom resolution	No significant effect of intervention on symptoms outcome in females. Delayed recovery time was noticed in males, compared with the control group. Participants tended to recover within the expected time frame.

TABLE 1 Continued

Source	Quality Rating	Study Design	Population			Intervention Group	Control Group	Outcomes	Results
			Intervention	Control					
Willer et al, 2019 ⁴⁵	ROBINS-I: moderate	Quasi-experimental trial	$n = 52$; mean age (SD), 15.3 (1.6) y; age range, 13–18 y; females, 24; males, 28	Rest ($n = 48$): mean age (SD), 15.4 (1.4) y; females, 12; males, 36. Placebo ($n = 51$): mean age (SD), 15.4 (1.7) y; females, 24; males, 27		An aerobic exercise intervention. Activity type, stationary bike or treadmill at home or a gym setting; frequency, daily for 4 wk or until recovery, whichever comes first; duration, 20 min, with a 5 min warmup and 5 min cool-down; intensity, 80% of HR causing exacerbated symptoms	Rest or placebo Rest: sports participation and exercise were prohibited. Participants were told to limit symptom exacerbators (ie, TV). Placebo: progressive stretching program. Activity type, breathing exercises and whole body stretches; frequency: daily; duration, 20 min; intensity, increased weekly	D until symptom resolution; PCS-R	The activity group showed improved, quicker recovery compared with the relative rest group. The relative rest group showed similar effects to the placebo group.

BDI-II, Beck Depression Inventory-II; BESS, Balance Error Scoring System; COPM, Canadian Occupational Performance Measure; f, female; impact, Immediate Post-Concussion Assessment and Cognitive Test; HR, heart rate; IQR, interquartile range; m, male; max, maximum; N/A, not applicable; Nall, physiologic + cervicogenic + vestibular treatment; Nc, Ncervicogenic; Ncv, cervicogenic + vestibular treatment; Np, Nphysiologic; Npc, physiologic + cervicogenic treatment; Npv, physiologic + vestibular treatment; Nv, Nvestibular; PCS, Post-Concussion Scale; PCSI, Post-Concussion Symptom Inventory; PCS-R, Post-Concussion Scale-Revised; PCSS, Post-Concussion Symptom Scale; ROB, Risk of Bias; SCAT-2, Sports Concussion Assessment Tool-2.

^a Age ranges are for all participants (intervention + control) combined.

On the basis of the heterogeneity assessment, combination of the studies within the outcome measures were appropriate to combine. However, there did appear to be clinical and statistical heterogeneity. Clinical heterogeneity appeared in the form of differing interventions and initiation time of intervention after injury. Therefore, a random effects model was used for the meta-analysis.

Risk of Bias

The risk of bias in included studies was mixed. Most RCTs had a high level of biases according to the Cochrane Risk of Bias scale for RCTs, primarily because of the inability to blind the participants to their group assignment given the nature of the intervention. The non-RCTs had moderate scores of bias as assessed through the ROBINS-I (Table 1).

Effect of Physical Activity on Resolution of Symptoms

Symptom-reporting was the most common outcome in the included studies ($n = 21$), using measures including the CogSport Symptom Scale, Post-Concussion Symptom Scale, Post-Concussion Scale-Revised, Health Behavior Inventory, Sports Concussion Assessment Tool 2, Sports Concussion Assessment Tool 3, and Post-Concussion Symptom Inventory. Most studies ($n = 13$) reported a decrease in number of symptoms after engagement in the activity-related intervention.

A meta-analysis was conducted with 7 RCTs that reported on symptom resolution. Activity engagement had a significant effect on symptom resolution (standardized MD, 0.39 [95% CI, 0.15 to 0.63]; I^2 , 0%; $P = .002$), illustrated in Fig 2.

Effect of Physical Activity on Quality of Life

QoL was assessed in 5 studies, 4 of which used the Pediatric Quality of

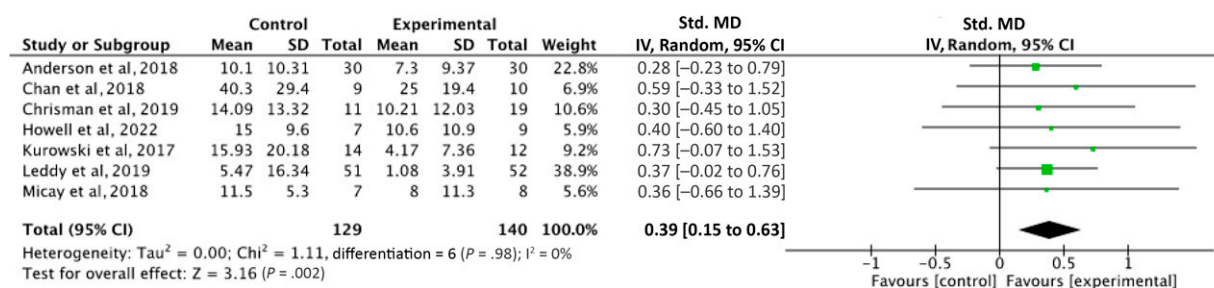


FIGURE 2
Forest plot for the effectiveness of activity-based interventions on symptoms.

Life Inventory (PedsQL) and 1 used the Children's Assessment of Participation and Enjoyment. QoL was operationalized as mental and physical health, school functioning, and activity participation. For the studies which used the PedsQL, half of the studies (50%) did not report a significant effect of activity engagement on QoL, whereas the other half showed statistical significance. For the 2 studies using the PedsQL assessing both child and parents, parents reported a larger improvement on QoL after the activity-based intervention, whereas children reported less improvement after the activity-based intervention, compared with participants receiving the control intervention.

A meta-analysis was conducted on 2 RCTs, both of which used the PedsQL. Engagement in activity did not have a significant effect on QoL (MD, -0.91 [95% CI, -7.76 to 5.94]; I^2 , 0%; $P = .79$), as shown in Fig 3.

Effect of Physical Activity on Days to Preinjury Activity Levels

Return to preinjury activity levels was measured in 5 studies. Included

studies operationalized return to activity by measuring the number of days from injury to a full return to play ($n = 4$) and using the Godin Leisure-time Exercise Scale ($n = 1$). Some studies ($n = 2$) reported a significant reduction in number of days to return to activity ($P < .05$) after the intervention, whereas other studies ($n = 2$) did not.

A meta-analysis was not conducted because only 1 RCT was identified. The RCT did not demonstrate significant results ($P = .82$) between the control (usual care) and the intervention (exercise) group on time until approval of return to a full activity.⁴³ Acute symptom severity measured by self-report on the Post-Concussion Symptom Scale was associated with days until return to full activity was found ($P = .04$); participants with a higher acute symptom severity had more days until return to activity.⁴³

DISCUSSION

This systematic review and meta-analysis aimed to examine the effects of social and physical activity

on symptoms, QoL, and days until recovery. Twenty-four studies were included in the review, of which 9 studies were included in the meta-analysis. Engagement in a physical activity-related intervention significantly reduced number and severity of symptoms, but did not significantly improve QoL.

Our review identified a reduction in symptom reporting in participants in the activity-based intervention groups. This aligns with previous literature which states that moderate activity inclusion soon after a concussion positively facilitates recovery.^{25,44,45} Notably, return to activity guidelines after concussion presents a clear path for a safe return to activity. When following guidelines, individuals slowly progress the level of activity participation depending on the prevalence of worsening or new emerging symptoms.¹² This is important in the youth population because they are reported to have longer recovery times compared with adults.^{44,46,47} A majority of the study interventions focused on aerobic-based activities, which were

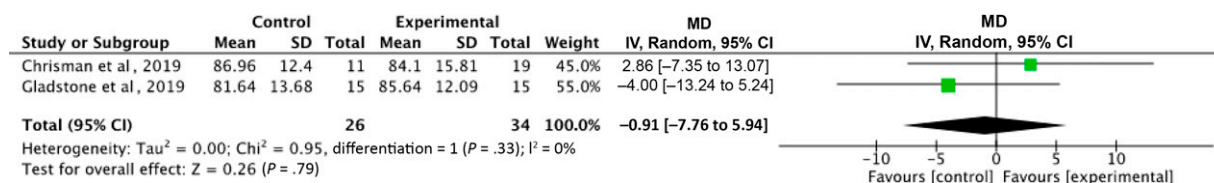


FIGURE 3
Forest plot of activity-based interventions on QoL outcomes.

undertaken by the participants on a daily basis. Interventions with a multicomponent aspect, such as the inclusion of visualization techniques or a mixture of coordination exercises and aerobic exercises, specifically showed improvements in symptoms. As such, providing activity-based interventions may reduce symptom burden and improve recovery.

Our study showed no significant improvement in QoL among groups after receiving physical activity-related interventions. QoL characteristics that were important to this review focused on physical and mental health and school life, as well as amount and type of participation in activities. Although slight improvements in QoL were present, a significant difference between intervention and control groups was lacking.²⁷ Gladstone et al³⁴ noted that QoL was reported to increase in both the control and intervention groups, resulting in no significant between-group difference. Therefore, this needs to be further studied to better inform guidelines for improved QoL. Specifically, it is important to note that social activity interventions may play a role, but could not be investigated empirically in this review.

Our review did not identify studies that included interventions with a sole component of social activity engagement. Social activity is a large component in the role and occupation of youth and can help recovery in other contexts and aid in improving psychological symptoms.⁴⁸ Specifically, it has been shown that the connection between social and physical activity can better mediate and decrease depression-related symptoms.^{48,49} Therefore, there is a known importance of social activity, and future studies should be

implemented to view its effect on QoL after a concussion.

When assessing returning to preinjury activity levels, our study did not include a meta-analysis because of a lack of RCTs. Assessing postinjury activity levels and comparing to preinjury levels heavily relies on one's perception and may hinder the results because of biases.⁵⁰ Return to activity can be influenced by several factors, including symptom severity, activity initiation, and sex. For example, recent literature has shown a relationship between symptom severity and days until full clearance is achieved, with greater symptom severity reports predicting increased days of activity restrictions.⁴² Additionally, a greater number of days until full return to activity was associated with delays in activity initiation after the injury event.^{38,43} Lastly, sex plays a factor in return to activity after concussion. Exercise resulted in males experiencing greater days until recovery, although both males and females recovered within the expected time frames.²¹ Future studies should implement RCTs to understand the effect of activity on days until full return to activity. In addition, future studies will need to account for sex and gender differences, concussion history, and symptom severity for an enhanced understanding.

Future research is needed to further enhance our understanding of the effect of activity on recovery. Social activity interventions could help promote recovery in youth by significantly improving one's QoL. Social support within a group-context can be important in the recovery experience of youth with concussion.¹⁵ Common themes of isolation and accommodation can be addressed by a group of trusted individuals, such as peers.¹⁵ Interventions including a combination of physical and social

activities should be completed to assess any potential benefits of both domains. Future studies should also determine how to ensure guidelines of activity are being adhered to after a concussion in youth. Adherence to guidelines occur mainly through self-management; however, symptoms related to concussion, such as depression, could hinder motivation and self-reliance.^{51,52} Aspects to improve adherence and motivation could be essential to maximize recovery, and hence, should be explored in future studies.

Limitations

Four main limitations to this study have been highlighted. First, there was only 1 RCT that measured days until return to activity, and as such, no meta-analysis was completed. This limits the understanding of how length of time until returning to activity can be influenced by activity-based interventions. However, the summary of the study in context of the other findings in the review provides insight for future research. Second, there was variation in the studies with regard to intervention initiation from time since injury, with some studies having a short time since injury (eg, days) and others having much longer time (eg, years). This may have impacted the precision of the findings. However, understanding the influence of interventions on key outcomes, not dependent on time since injury, provides an understanding of the global benefit of the interventions. Third, 13 studies were rated as having high or serious risk of bias. Bias in these studies strongly relate to a difficulty in participants being blinded to their intervention and the outcomes. This was difficult because of the nature of the intervention (physical activity versus rest/normal care). Additionally, many outcomes were assessed through self-report questionnaires, making it difficult to

conceal. Fourth, we included studies with adults as long as individuals aged <18 years made up a majority (>50%) of the included population. Four studies included mixed populations which could affect the generalizability of our findings. This could have hindered our findings because adults and children see significant differences in their recovery.

CONCLUSIONS

Engagement in activity can produce many health benefits. In this review, activity-related interventions for youth with concussion can provide positive effects, including lower symptom reporting. Future research and implementation methods of activity-based interventions, specifically social activities, need to be evaluated in this population.

ABBREVIATIONS

CI: confidence interval
MD: mean difference
PedsQL: Pediatric Quality of Life Inventory
QoL: quality of life
RCT: randomized controlled trial
ROBINS-I: Risk of Bias in Nonrandomized Studies of Intervention

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